Project: Smart Brewery

Description:

The Smart Brewery Project aims to modernize home brewing through IoT and real-time analytics. By utilizing smart sensors for brewing temperature, fermentation, and ingredient control, it enhances brew quality and flavor consistency. Home brewers can automatically manage their brewing operations, receive alerts, access insights about their brewing recipe history files, and receive AI recipe suggestions. This project represents a fusion of traditional craft with data-driven efficiency, promising a transformative impact on the home brewing industry by improving the brewing process and overall experience

Use Cases:

1. **Brew a recipe:** Users can customize personalized brewing recipes, and with a simple click watch as the system autonomously executes each step. This feature provides the user easy control, turning home brewing into a dynamic, hands-free experience for enthusiasts of all levels. The users can elevate their brewing journey with the perfect blend of precision and convenience, as every crafted recipe comes to life effortlessly with our intuitive system.
2. **View brewing history:** For those who seek continuous improvement, the product provides a Batch History and Analytics feature. Keep track of each brewing session, including all relevant data. This feature is perfect for users who want to search their brewing history, allowing them to enhance their memory of past beers and revisit the flavors they achieved in previous brewing sessions. This capability transforms each brewing experience into a valuable learning opportunity.
3. **Get AI beer recipes:** Explore AI-crafted beer recipes customized to diverse tastes and styles, providing inspiration for your brewing desires. Receive personalized recommendations based on your preferences, ensuring a unique and enjoyable brewing experience.
4. **Share & download recipes online:** Engage with a brewing community as you share your unique recipes online, fostering collaboration and knowledge exchange. Explore an extensive collection of user-generated recipes, offering a diverse array of brewing styles and techniques to download and experiment with. Connect with fellow brewers, contributing to the collective evolution of homebrewing through seamless recipe sharing and downloading.

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Requirements:

General:

- **Sync Feature:** Enable effortless synchronization between the app and the heat unit's temperature sensor, as well as the fermentation sensor, using Bluetooth or WiFi connectivity. This ensures real-time data exchange, allowing the app to automatically update the cloud based server with the latest temperature and fermentation information gathered from the connected devices. Users can thus easily monitor and control brewing conditions remotely, fostering a user-friendly and efficient brewing experience.
- **User Accounts Management feature:** Users need to have individual accounts to access and save personalized brewing recipes.
  Functions:
    1. Create new accounts
    2. Update account data
    3. Delete accounts
- **Notification Feature:** Receive instant updates on your brewing process with the notification feature, alerting you to crucial milestones, temperature adjustments, and fermentation progress, ensuring you stay informed and in control of every aspect of your beer-making journey.
- **Performance:** The app should deliver optimal performance, providing users with swift access to real-time brewing data, intuitive controls, and seamless synchronization with brewing devices, ensuring a smooth and efficient homebrewing experience.
- **Security:** The app should prioritize security by implementing robust encryption protocols to safeguard sensitive brewing data, ensuring a secure and confidential environment for users to manage their home brewing processes.
- **Reliability:** The app should be reliable and stable, experiencing minimal crashes or bugs. Additionally, it should have the capability to recover from errors and provide helpful messages to users.

Brew a Recipe:

- **Recipe Input:** A user-friendly interface for users to input and customize brewing recipes.
  Functions:
  1. Create new recipe
  2. Update an existing recipe
  3. Delete recipe
- **Automated System:** Integration of an iot automated brewing system monitoring temperature sensor and controlling heat unit, capable of executing recipes with a single click.
- **Control Interface:** An intuitive control interface to provide users with easy control over the brewing process.
  Functions:
  1. Show the recipe stage
  2. Show the current temperature
  3. Show the remaining time of the current stage/entire process
  4. Show fermentation data

Brewing History:

- **Data Logging:** Implement a feature for the system to log relevant data from each brewing session into a database.
- **Search Functionality:** A search functionality for users to efficiently locate and review specific brewing sessions, sorted by name, oldest/newest, rate, etc.
- **Analytics Module:** Integration of an analytics module to provide insights and trends from brewing history.
- **Brewing Taste Result:** Establish a mechanism for users to input and update their brewing flavors result.

Get AI Beer Recipes:

- **AI Integration:** Implement an artificial intelligence system capable of crafting diverse and customized beer recipes by customized parameters.
- **Data inputs:** Using the user's brewing history and taste rates results data to run the AI engine for beer recipes recommendations.

Share & Download Recipes Online:

- **User Profiles:** Enable user profiles for individuals to contribute and share their recipes to their followers who can download those recipes to their "favorites".
- **User Contribution:** Allow users to contribute, rate, and review recipes, fostering a collaborative community.